

A complete inverse for all Golay-octad traces

Use the standard projective Leech model $\lambda = x/\sqrt{8}$. For each of the 759 Golay octads B , record

$$a_B = \frac{1}{4} \sum_{i \in B} x_i, \quad S = \sum_B a_B, \quad S_i = \sum_{B \ni i} a_B.$$

Every coordinate returns.

$$x_i = \frac{23S_i - 7S}{1012}.$$

For integer data, first require the explicit linear image equations $a_B = \frac{1}{4} \sum_{i \in B} x_i(a)$. There is then exactly one remaining integral obstruction:

$$\frac{11S - 23S_0}{506} \equiv 0 \pmod{4}.$$

This condition is necessary and sufficient for a Leech lift. The lift is unique.

Why these constants? The octad incidence matrix satisfies

$$A^\top A = 176I + 771\mathbf{1}^\top.$$

The positive-octad vectors generate an index-four sublattice. Its dual is $\Lambda + \mathbb{Z}\mathbf{1}/(2\sqrt{8})$, yielding the exact syndrome above.

Why retain all the data? A previous nine-trace time readout is not stable under the second symmetry generator. The joint symmetry orbit supplies all octads, and their linear span has rank 24. The inverse restores the data that the smaller quotient loses.

Full statement: Symmetry-complete integral trace reconstruction for the projective Leech lattice.